

Problem Set 2 – Set identities

Session 2 · Sets & set operations

Part A · Notation and computation

1 NOTATION

Let $U = \{1, 2, \dots, 12\}$, $A = \{x \in U : x \text{ is even}\}$ and $B = \{x \in U : x \text{ is a multiple of } 3\}$.

- Write A and B in roster notation.
- Compute $A \cap B$, $A \cup B$, $A \setminus B$, $A \triangle B$ and B^c .

2 $\in$ VS $\subseteq$

Let $C = \{1, \{1\}, \{1, 2\}, \emptyset\}$. Decide true or false, and in each case quote the definition that settles it.

- $1 \in C$
- $\{1\} \in C$
- $\{1\} \subseteq C$
- $\{1, 2\} \subseteq C$
- $\emptyset \subseteq C$
- $|C| = 4$

3 POWER SET

Let $A = \{a, b, c\}$.

- List $\mathcal{P}(A)$ in full, organised by the size of the subset.
- State $|\mathcal{P}(A)|$ and explain in one sentence why $|\mathcal{P}(A)| = 2^{|A|}$ for *any* finite A .

Part B · Six identities to prove

4 DISTRIBUTIVITY

Prove that $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ for all sets A, B, C .

5 DE MORGAN

Let $A, B \subseteq U$. Prove that $(A \cap B)^c = A^c \cup B^c$.

6 DIFFERENCE

Prove that $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$ for all sets A, B, C .

7 SYMMETRIC DIFFERENCE

Prove that $A \triangle B = (A \cup B) \setminus (A \cap B)$ for all sets A, B .

8

SUBSETS

Prove that for all sets A, B : $A \subseteq B \iff A \cup B = B$.

9

CARTESIAN PRODUCT

Prove that $A \times (B \cap C) = (A \times B) \cap (A \times C)$ for all sets A, B, C .